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### **SYSTEMS AND METHODS FOR REMOTELY MONITORING ELECTRONIC DISPLAYS**

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#### **Abstract**

Systems and methods for remotely monitoring digital signage are disclosed. A display assembly has components including a cooling fan, an electronic display, and a sensor which monitors, in substantially real time, an operating condition at the display assembly. A programmable controller is in electronic communication, by way of the internet, with the sensor and at least one of the components of the display assembly. A remote computer receives and stores, in substantially real time, data representative of actual performance of the operating condition from the sensor, automatically analyzes the actual performance data in comparison with preferred performance data for the operating condition stored at a database to determine if they differ by at least a preset threshold difference, and if so, automatically sends an electronic signal to the controller to adjust the actual performance of the operating condition to align with the preferred performance condition setting.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. application Ser. No. 18/935,184 filed Nov. 1, 2024, which is a continuation of U.S. application Ser. No. 18/635,228 filed Apr. 15, 2024, now U.S. Pat. No. 12,169,658 issued Dec. 17, 2024, which is a continuation of U.S. application Ser. No. 18/125,694 filed Mar. 23, 2023, now U.S. Pat. No. 11,989,476 issued May 21, 2024, which is a continuation of U.S. application Ser. No. 17/974,361 filed Oct. 26, 2022, now U.S. Pat. No. 11,645,029 issued May 9, 2023, which is a continuation-in-part of U.S. application Ser. No. 17/893,253 filed Aug. 23, 2022, now U.S. Pat. No. 11,614,911 issued Mar. 28, 2023, which is a continuation of U.S. application Ser. No. 17/548,698 filed Dec. 13, 2021, now U.S. Pat. No. 11,455,138 issued Sep. 27, 2022, which is a continuation of U.S. application Ser. No. 17/134,634 filed Dec. 28, 2020, now U.S. Pat. No. 11,243,733 issued Feb. 8, 2022, which is a continuation of U.S. application Ser. No. 16/508,877 filed Jul. 11, 2019, now U.S. Pat. No. 10,908,863 issued Feb. 2, 2021, which claims the benefit of U.S. Provisional Application No. 62/697,016 filed Jul. 12, 2018, the disclosure of each of the foregoing are hereby incorporated by reference as if fully restated herein.

## TECHNICAL FIELD

[0002] Exemplary embodiments relate generally to systems and methods for remotely monitoring electronic display assemblies.

## BACKGROUND AND SUMMARY OF THE INVENTION

[0003] The use of electronic displays has increased in recent years. Beyond personal use, these electronic displays are sometimes placed in an enclosure, which may be ruggedized, for both indoor and/or outdoor use. For example, without limitation, such display assemblies may be placed outdoors, such as on sidewalks or on the tops of vehicles to name some examples, to display advertisements or other information. Oftentimes these display assemblies are associated with other electronic equipment to provide features such as, but not limited to, video conferencing, web browsing, way finding, image capture, emergency notification, and the like. It is desirable to provide owners, manufactures, operators, renters, or the like of such display assemblies with access

to operations data regarding the status and operation of such display assemblies. Furthermore, these display assemblies consume a significant amount of power during operation. A significant amount of the energy consumption of such display assemblies comes from operation of the backlight. Particularly as the use of such display assemblies has increased, it is desirable that such operations data include data regarding the energy consumption of the display, including but not limited to, status and operation of the backlight. Such operations data may be stored and presented to display owners and other relevant parties such that energy consumption and reliability may be tracked. Certain operations parameters may be adjusted in response to the presented data. For example, without limitation, backlight levels may be adjusted downward to reduce energy consumption or maintenance frequencies may be increased in response to reported display assembly downtime. [0004] However, for many such display assemblies, multiple parties may have an interest in the operations data. For example, a display assembly manufacturer may have an interest in monitoring its tens, hundreds, thousands, tens of thousands, etc. of display assemblies. Each owner or renter of one or more such display assemblies might likewise have an interest in monitoring the display assemblies it owns. While the manufacturer may already have access to this operations data, the operations data may include data for all owners or renters. Thus, the operations data may be co-located with operations data for other owners or renters. Allowing complete access to such data may create privacy concerns. However, individual storage of each client's information would potentially require multiple storage devices, increasing complexity and expense. Therefore, what is needed is a system and method for providing access to co-located operations data for a display assembly.

[0005] The present disclosures provide a system and method for providing access to co-located operations data for a display assembly. The system may comprise one or more client devices in communication with a centralized monitoring center via a network. Multiple display assemblies may likewise be in communication with the monitoring center via the network. In exemplary embodiments, each display assembly comprises one or more electronic displays and other electronic components for operation of the display assembly in electrical connection with a display controller. The display controller may likewise be in electrical connection with a network interface device. The network interface device may likewise be in communication with the centralized monitoring center via the network.

[0006] Operations data may be continually gathered and transmitted to the central monitoring center for storage. The received operations data may be processed. Such processing may determine which client(s) are associated with the operations data and what the operations data represents. The operations data may be summarized and presented in a visual depiction. Upon receipt of a client request for operations information, client specific information may be retrieved and presented to the client in the visual depiction. In other exemplary embodiments, upon login, a client may be permitted to remotely operate or modify display assemblies associated with the client.

[0007] Remote monitoring and/or operation of display assemblies may permit display owners, operators, users, combinations thereof, or the like to monitor display assembly status and/or exert operational control over the same from various locations. A wide variety of data points, which may include information, configuration settings, operational parameters, combinations thereof, or the like related to status and/or operation of the display assemblies may be monitored and/or updated.

[0008] Monitored data points may include, for example without limitation, one or more of: internal temperatures, external temperatures, ambient weather conditions, ambient lighting, backlight levels, ambient pressures, internal pressures, geographic locations, power consumption information, power input information, power output information, status information, identifying information, proof of play information, humidity information, configuration information, usage information, fan speed information, combinations thereof, or the like. Such settings may include, for example without limitation, one or more of: fan speed, backlight levels, day-to-night or night-to-day transition information, temperature tolerances, dewpoint spread tolerances, combinations

thereof, or the like. Any number, type, and/or kind of settings and/or data points may be tracked, reported, and/or updated. Any number, type, and/or kind of sensors may be provided for determining the various data points, such as but not limited to temperature sensors, pressure sensors, ambient light sensors, airspeed sensors, power level sensors, combinations thereof, or the like. The data points and/or settings may be filtered for more focused presentation, such as but not limited to, based on customer identifier, geographic region, part number, serial number, type of metric or setting, applicable dates, applicable ranges, combinations thereof, or the like. The data points may be presented in summary form, such as for a single display assembly and/or multiple display assemblies, over time (e.g., in time dependent graphical form), table form, spreadsheet, in dashboard form, or in detailed form for a single or multiple display assemblies. The data may be presented for a single display or multiple displays. Displayed data may be limited to that associated with a particular customer identifier, in exemplary embodiments, without limitation.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

[0009] In addition to the features mentioned above, other aspects of the present invention will be readily apparent from the following descriptions of the drawings and exemplary embodiments, wherein like reference numerals across the several views refer to identical or equivalent features, and wherein:

[0010] FIG. 1 is a simplified plan view of an exemplary system;

[0011] FIG. 2 is a detailed view of an exemplary display assembly of FIG. 1;

[0012] FIG. 3 is a detailed view of an exemplary monitoring center of FIG. 1;

[0013] FIG. 4 is a detailed view of an exemplary client device of FIG. 1;

[0014] FIG. 5 is a flow chart of exemplary logic for use with the system of FIG. 1;

[0015] FIG. 6 is a flow chart of other exemplary logic for use with the system of FIG. 1;

[0016] FIG. 7 is an exemplary user interface for use with the system of FIG. 1;

[0017] FIG. 8 is another exemplary user interface for use with the system of FIG. 1;

[0018] FIG. 9 is another exemplary user interface for use with the system of FIG. 1;

[0019] FIG. 10 is another exemplary user interface for use with the system of FIG. 1;

[0020] FIG. 11 is another exemplary user interface for use with the system of FIG. 1;

[0021] FIG. 12 is another exemplary user display generated by at least the system of FIG. 1;

[0022] FIG. 13 is another exemplary user display generated by at least the system of FIG. 1;

[0023] FIG. 14 is a continuation of the user display of FIG. 13;

[0024] FIG. 15 is a continuation of the user display of FIG. 14;

[0025] FIG. 16 is a continuation of the user display of FIG. 15;

[0026] FIG. 17 is another exemplary user display generated by at least the system of FIG. 1;

[0027] FIG. 18 is another exemplary user display(s) generated by at least the system of FIG. 1;

[0028] FIG. 19 is another exemplary user display(s) generated by at least the system of FIG. 1;

[0029] FIG. 20 is another exemplary user display generated by at least the system of FIG. 1; and

[0030] FIG. 21 is another exemplary user display generated by at least the system of FIG. 1.

### **DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENT(S)**

[0031] Various embodiments of the present invention will now be described in detail with reference to the accompanying drawings. In the following description, specific details such as detailed configuration and components are merely provided to assist the overall understanding of these embodiments of the present invention. Therefore, it should be apparent to those skilled in the art that various changes and modifications of the embodiments described herein can be made without departing from the scope and spirit of the present invention. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

[0032] Embodiments of the invention are described herein with reference to illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the invention should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

[0033] FIG. 1 is a simplified block diagram of an exemplary system. A number of display assemblies 30 may be in communication with a monitoring center 20. Similarly, a number of client devices 10 may be in communication with the monitoring center 20. The display assemblies 30 may be located remote from the monitoring center 20. Likewise, the client devices 10 may be located remote from the monitoring center 20. The communication between the display assemblies 30, the monitoring center 20, and the client devices 10 may be made by way of a network 80. The network 80 may be any network such as a cellular network, internet, intranet, world wide web, or the like. The network 80 between the client devices 10 and the monitoring center 20 may be the same or may be different from the network 80 between the monitoring center 20 and the display assemblies 30.

[0034] Each of the display assemblies 30 may be any kind of display assembly 30, such as but not limited to, a free-standing display kiosk configured for placement on a sidewalk, a wall-mounted display unit, a vehicle toppler unit, or the like. Each of the client devices 10 may be a personal electronic device such as, but not limited to, a smartphone, tablet, smartwatch, laptop, desktop computer, some combination thereof, or the like.

[0035] FIG. 2 is a detailed view of an exemplary display assembly 30 of FIG. 1. Each display assembly 30 may comprise one or more electronic displays 70 in electrical connection with a display controller 50. Each electronic display 70 may comprise a backlight for illuminating the electronic display 70. The electronic display 70 may comprise one or more a liquid crystal displays, light emitting diode (“LED”) displays, organic LED displays, plasma displays, some combination thereof, or the like. One or more of the electronic displays 70 may comprise touch capabilities. The backlight may be comprised of a number of LEDs arranged in a directly backlit, edge-lit, or other orientation.

[0036] In exemplary embodiments, the display controller 50 may be in electrical connection with other components 60 for operating the display assembly 30. Such components 60 may include, but are not limited to, fans, temperature sensors, light sensors, fan speed sensors, power consumption sensors, air quality sensors, weather sensors, other sensors, telephone equipment, video conferencing equipment, voice over internet protocol (VOIP) equipment, touch screens, camera, microphones, emergency notification devices, processors, electronic storage devices, wayfinding equipment, location detection devices, video players, proof of play devices, and the like. Any number, combination, and/or type of components are contemplated.

[0037] The display controller 50 may be configured to gather operations data from the electronic display 70. The display controller 50 may, alternatively or additionally, be configured to gather operations data from the components 60. Such operations data may include, but is not limited to, proof of play data, fan speed data, temperature data, humidity data, power consumption data, ambient light data, weather data, backlight data, electronic display data, status data, emergency notification data, air quality data, sensor readings, camera images or video recordings, microphone audio recordings, use history, and the like. The operations data for multiple clients may be co-located at the electronic storage device 52 located at the display assembly 30. Portions of the operations data may be associated with one or more client identifiers. For example, without limitation, operations data associated with a particular image displayed on the electronic display 70 may be associated with a particular client identifier. Alternatively, or in addition, all operations data originating from one or more display assemblies 30 may be associated with one or more client identifiers. This may permit for specific clients to access and retrieve only the operations data

associated with the particular client-permitting the co-location of such operations data while maintaining privacy. The association with a client identifier may be performed at the display assembly **30** or at the monitoring center **20**. The client identifiers may be, without limitation, unique alphanumeric serial numbers.

[0038] The display controller **50** may be in electrical communication with a network connection device **40**. The network connection device **40** may be configured to transmit information to the monitoring center **20**, including but not limited to, the operations data. In exemplary embodiments, the network connection device **40** may likewise be configured to receive information from the monitoring center **20**, including but not limited to, operation instructions. Such operations instructions may comprise remote login capabilities, remote viewing, fan speed instructions, backlight adjustment instructions, video or image files for display on the electronic displays **70**, some combination thereof, or other instructions for operation of the display assembly **30**. The network connection device **40** may transmit and/or receive such information by way of the network **80**.

[0039] The display controller **50** may comprise an electronic storage device **52** for storing the operations data and/or operations instructions. The display controller **50** may also comprise a processor **54** for processing the operations data and/or operations instructions. The electronic storage device **52** may comprise software instructions, which when executed, configure the processor **54** to perform various steps and processes described herein.

[0040] FIG. **3** is a detailed view of an exemplary monitoring center **20** of FIG. **1**. The monitoring center **20** may comprise an electronic storage device **52** for storing the operations data and/or operations instructions. The monitoring center **20** may also comprise a processor **54** for processing the operations data and/or operations instructions. The electronic storage device **52** may comprise software instructions, which when executed, configure the processor **54** to perform various steps and processes described herein. The processor **54** may be the same or different from the processor **54** of the display assembly **30**.

[0041] The monitoring center **20** may further comprise a network connection device **40**. The network connection device **40** may be in electrical communication with the processor **54** and the electronic storage device **52**. The network connection device **40** may be in electrical communication with one or more of the display assemblies **30**. The network connection device **40** may be configured to receive operations data from the one or more display assemblies **30**. The network connection device **40** may also be configured to transmit operations instructions to one or more of the display assemblies **30**.

[0042] The network connection device **40** may be receive the operations data and pass it to the electronic storage device **52** for storage. The operations data for multiple display assemblies **30** may be co-located at the electronic storage device **52** located at the monitoring center **20**. Co-location may be accomplished by associating portions of the operations data with one or more client identifiers. Each portion of the operations data associated with a particular client identifier may be stored on a common electronic storage device **52**, but electronically partitioned to ensure that the operations data associated with each particular client is maintained separate.

[0043] Alternatively, or in addition, all operations data originating from one or more display assemblies **30** may be associated with one or more client identifiers. This may permit for specific clients to access and retrieve only the operations data associated with the particular client-permitting the co-location of such operations data while maintaining privacy. The network connection device **40** may transmit and/or receive such operations data and/or operations instructions by way of the network **80**.

[0044] The gathering, transmitting, storing, receiving, and retrieving of operations data and/or operations instructions as shown and described herein may be accomplished, wholly or in part, by the use of a microservices architecture. Any of the other steps or methods described herein may likewise be accomplished, wholly or in part, by the use of a microservices architecture.

[0045] The monitoring center **20** may be a brick-and-mortar location staffed with a number of monitoring personnel, though such is not required. In exemplary embodiments, the monitoring center **20** may comprise one or more rooms with one or more displays which may provide status and/or operations information for one or more display assemblies **30**. For example, without limitation, a complete or partial list of display assemblies **30** and status information for each display assembly **30** may be shown. Information for a subset of display assemblies **30** may be shown in a rotating, scrolling, or other fashion. Detailed information regarding one or more display assemblies **30** may be shown on the same or a separate screen. A number of personal electronic devices, such as but not limited to, smartphone, tablet, smartwatch, laptop, desktop computer, some combination thereof, or the like may likewise display some or all of the information and may be configured to receive user input comprising operational instructions for one or more of the display assemblies **30**.

[0046] FIG. **4** is a detailed view of an exemplary client device **10** of FIG. **1**. The client device **10** may be any personal electronic device, including but not limited to, a smartphone, tablet, smart watch, laptop, desktop computer, some combination thereof, or the like. The client device **10** may comprise an electronic storage device **52** for storing the operations data and/or operations instructions. The client device **10** may also comprise a processor **54** for processing the operations data and/or operations instructions. The electronic storage device **52** may comprise software instructions, which when executed, configure the processor **54** to perform various steps and processes described herein. The processor **54** may be the same or different from the processor **54** of the client device **10**.

[0047] The client device **10** may further comprise a network connection device **40**. The network connection device **40** may be in electrical communication with the processor **54** and the electronic storage device **52**. The network connection device **40** may be in electrical communication with the monitoring center **20**. The network connection device **40** may be configured to receive operations data from the one or more display assemblies **30** by way of the monitoring center **20**. In exemplary embodiments, the network connection device **40** may be receive such operations data and store it on the electronic storage device **52**. The network connection device **40** may be configured to transmit operations instructions to the monitoring center **20**. The network connection device **40** may transmit and/or receive such information by way of the network **80**. In exemplary embodiments, the receipt of operations data and/or transmission of operations instructions is accomplished by use of an internet browsing application and an internet-based user interface **90**.

[0048] FIG. **5** is a flow chart of exemplary logic for use with the system of FIG. **1**. The display controller **50** may gather operations data the one or more electronic displays **70** and/or the other components **60** of the respective display assembly **30**. The operations data may be associated with one or more client identifiers. This operations data may be stored on the electronic storage device **52** of the respective display assembly **30**. The operations data may be transmitted by way of the network connection device **40** and the network **80** to the monitoring center **20** where it may be stored on one or more electronic storage devices **52** at the monitoring center **20**. In exemplary embodiments, the processor **54** at the monitoring center **20** may associate the operations data with one or more client identifiers. The processor **54** may electronically partition the electronic storage device **52** such that operations data associated with each particular client identifier is kept separate from operations data associated with other client identifiers. This process may be repeated continuously or at any interval. Alternatively, or in addition, a client identifier may be associated with the operations data when transmitted to the monitoring center **20**.

[0049] A client request for operations data may be received from one or more of the client devices **10** at the monitoring center **20**. The monitoring center **20** may retrieve the operations data associated with the client. In exemplary embodiments, this may involve retrieving all operations data associated with one or more of the display assemblies **30** associated with the client. Alternatively, or in addition, operations data specific to the client from a particular display

assembly **30** may be retrieved. Stated another way, any single display assembly **30** may comprise operations data for multiple clients and only the operations data specific to the particular client may be retrieved. The retrieved operations data specific to the client may then be transmitted to the appropriate client device(s) **10**. In exemplary embodiments, data specific to the client may be identified by way of the client identifiers. Operations data may be requested from each electronic display assembly **30** periodically, continuously, sequentially, in a particular order, some combination thereof, or the like.

[0050] FIG. **6** is a flow chart of other exemplary logic for use with the system of FIG. **1**. One or more display assemblies **30** may be associated with one or more clients. A client login request may be received by way of one or more client devices **10** at the monitoring center **20**. If the client login is not successful (e.g., if the user name and/or password are incorrect) the request may be denied. If the client login is successful (e.g., the user name and password are correct) the client may be permitted to remotely access one or more display assemblies **30** associated with the client. In exemplary embodiments, client login may be made by way of user name and password, one time use codes, biometric information, some combination thereof, or the like. Any method or system for verifying client identity is contemplated. Such access may be made by way of the monitoring center **20**. The client may be able to view and retrieve operations data for the one or more display assemblies **30** associated with the client. The client may also be permitted to submit operation instructions for the one or more display assemblies **30** associated with the client. The operations of the one or more display assemblies **30** associated with the client may then be modified to reflect the received client operation instructions. Permissions may be tailored on a per client basis such that only certain display assemblies **30** may be accessed and/or modified when particular client login information is provided.

[0051] FIG. **7** is an exemplary user interface **90** for use with the system of FIG. **1**. The user interface **90** may display overview information regarding a number of display assemblies **30**, each of which may be associated with a particular client though such is not required. Information about each display assembly **30** may include, but is not limited to, the serial number, description information, site information, and status information of each display assembly **30**. Status information, current and/or historical, may also be provided.

[0052] As shown in FIG. **8**, event information for each of the display assemblies **30** associated with a particular client may be depicted under an events tab **92**. The event information may include, but is not limited to, errors and warnings associated with one or more particular display assemblies **30**. An option to automatically generate a repair request ticket **94** may be provided for each event.

[0053] FIG. **9** is another exemplary user interface **90** for use with the system of FIG. **1**. A visualization **96** of current and historical operations data may be displayed. Such operations data may comprise data from one or more sensors associated with a display assembly **30**. In exemplary embodiments, the visualization **96** may be displayed in tabular form or a graphical representation such as, but not limited to, a line chart, bar chart, scatter diagram, or the like. Various categories of data may be depicted in the visualization **96** as selected by use of a menu **98**. Such categories may be displayed in a color-coded fashion.

[0054] FIG. **10** is an exemplary summary dashboard **91**, which may be configured to display summary information regarding a particular display assembly **30**. Environmental information **102** may be displayed such as, but not limited to, the average power supply temperature and average backlight temperature. Display status information **104** may also be displayed such as, but not limited to, the software version, the system run time, the backlight run time, and the brightness of each display. Additional information **106** may also be displayed such as, but not limited to, the status and resolution of various inputs, the inlet power supply, and the fan speed of various fans in the display assembly **30**. Any open repair request tickets may be displayed at an open ticket information area **108**. A screen capture **110** of what is currently being displayed, was previously displayed, and/or is scheduled to be displayed next on the display assembly **30** may also be



provided.

[0055] FIG. **11** is another exemplary summary dashboard **93**, which may be configured to display summary information regarding all display assemblies **30** associated with a particular client identifier. The number of display assemblies **30** with some downtime event may be depicted in a downtime visualization **112**. In exemplary embodiments, the downtime visualization **112** is a bar graph tracked by date, though any form of visualization is contemplated. A listing of top display assemblies with events **114** may be provided. A summary of repair ticket requests **116** may also be provided depicting all open repair ticket requests. A summary of the most reported events **118** may list the event(s) in question and the number of display assemblies **30** affected by the given event(s). The summary of the most reported events **118** may be presented in tabular form, though such is not required. Finally, a status summary **120** of all display assemblies **30** associated with a given client identifier may be provided. The status summary **120** may include, but is not limited to, the number of display assemblies **30** associated with the given client identifier which are playing video, not playing video, having no or limited network connectivity, in good condition, in critical condition, or having one or more warning events.

[0056] FIG. **12** through FIG. **21** illustrate exemplary user displays **90** that may be generated by the display assemblies **30**, systems, and/or methods shown and/or described herein. Some or all of the display assemblies **30** may comprise the components **60** and/or the controllers **50**, which may individually or collectively be configured to monitor, report, and/or adjust all of the data points shown and/or described. Such data points may be provided in raw, processed, and/or summarized form. The user displays **90** may comprise visualized data and/or provide options for a user to select, manipulate, filter, present, combinations thereof, or the like, the data points. The user displays **90** may present data, such as in a read-only format, and/or may provide options for the user to input new operational settings, parameters, filter criteria, combinations thereof, or the like.

[0057] The user displays **90** may comprise data specific to display assemblies **30** associated with a received customer identifier and/or other criteria, such as but not limited to geographic region, part number, serial number, combinations thereof, or the like. The user displays **90** may be generated in accordance with one or more user preferences or settings associated with various customer identifiers, by way of non-limiting example. The user displays **90** may be provided in various formats (e.g., read-only, read-write) based on permissions associated with various customer identifiers, by way of non-limiting example.

[0058] A number of exemplary user displays **90** are shown and/or described herein without limitation, such as without limitation as to the type, kind, and/or arrangement of data. Any amount, type, kind, and/or arrangement of data may be provided at any type, kind, and/or number of user displays **90**. The data may be presented in such user displays **90** in any type or kind of format, such as but not limited to table, spreadsheet, graphical (e.g., line chart, bar chart, plot, pie chart, combinations thereof, or the like), text, combinations thereof, or the like. Maximum, minimum, mean, median, mode, or the like values may be generated and/or provided. The data may be presented and updated in substantially real time or represent historical data and presented for particular time periods, by way of non-limiting example.

[0059] The user display **90** of FIG. **12** includes data for multiple display assemblies **30** and may include information such as, but not limited to, geographic position, status, total number of units, combinations thereof, of the like.

[0060] The user displays **90** of FIGS. **13-16** include data for a single display assembly **30**. Such data may be generated upon selection or inputting criteria for a particular display assembly **30**, such as but not limited to, serial number, customer identifier, geographic location, combinations thereof, or the like which is sufficient to specify a particular one of the display assemblies **30**. Such detailed information may be available by selecting one of the units described in a summary user display **90** such as but not limited to those shown and/or described with regard to FIG. **12**.

[0061] The user display **90** of FIG. **17** includes current displayed image information and touch

interface data, among other data points, for multiple ones of the display assemblies **30**. [0062] As illustrated in FIG. **18**, the user display **90** may include, for example without limitation, identifying information for display assemblies **30** such as but not limited to, part number, model number, IP address information, software version information, location information, city, state or province, unit serial number, country, uptime, contract, last poll time, IP address, update status, software revision, part number, model number, combinations thereof, and the like; status information, such as but not limited to open or closed status information for multiple ones of the electronic displays **70** or related subassemblies and/or access panels (e.g., doors, locks, etc.), magnet and/or latch status; power information, such as but not limited to system power information, including but not limited to direct current (DC) power information, surge, kW/H, efficiency, inputs (voltages, watts, amps; fan information, such as but not limited to fan status information, such as but not limited to fans associated with one or more open loop airflow pathways and/or closed loop airflow pathways, RPM speed, speed or power percentage, combination thereof, or the like; display or backlight information for multiple ones of the display assemblies **30**, such as but not limited to backlight status information, such as actual power, desired power, actual luminance, desired luminance, settings for various times of day, temperature information, combinations thereof, or the like; weather information local weather and/or internal relative humidity status information; environmental information; internal temperature information, such as for various components; location (e.g., GPS) information; touch screen status information (touch locations, touch count, frame numbers, temperature information, electrical status information); video status information (e.g., video state, video resolution, frame rate, connection information); display configuration information, such as board time, start up commands, screen information, video input information, test pattern, audio check, humidity check, humidity threshold, pressure check, location check, sensor type, shock sensitivity rating, frame capture orientation, combinations thereof, or the like; network configuration information, such as DHCP state, gateway information, net mask, DNS, server address, proxy address, URL, time zone, weather service, secondary IP, host name, combinations thereof, or the like; power configuration information, such as number of power supplies, min/max voltages, current, power, combinations thereof, or the like; day-to-night and/or night-to-day backlight transition confirmation information, such as time and backlight levels; temperature control configuration information, such as delta temperature change warning thresholds (e.g., ambient, internal), min/max temperature thresholds (e.g., ambient, internal), offsets; uptime statics and other information for multiple ones of the display assemblies **30**, such as but not limited to display information (e.g., serial number), geographic information (e.g., address information, site ID), runtime (e.g., percentage) information, play count information, play duration average information (e.g., in second), combinations thereof, and the like; combinations thereof, or the like. All such values or other information are merely exemplary and are not intended to be limiting. The values or other information may be presented in any of the normal associated units, or multiple such units (e.g., converted to/from metric to imperial). The values or other information may be presented for a single unit or multiple ones of the display assemblies **30** and/or various components thereof. The values or other information may be presented in a table, by unit serial number, in condensed or expanded form, summary form, combinations thereof, or the like.

[0063] FIG. **19** illustrates other exemplary user displays **90** comprising exemplary graphical presentations of monitored data points that may be generated. Such visualizations may be generated for particular time periods and/or on a substantially real-time basis, such as to account for time for data transmission and processing. For example, the user display **90** of FIG. **20** illustrates exemplary temperatures, exemplary temperature and fan information, exemplary power inlet information (e.g., alternating current (AC)), exemplary backlight drive and cloud cover information, exemplary power supply (e.g., DC voltages and/or currents), exemplary power input and output information, exemplary humidity information (e.g., dewpoint humidity pressure), exemplary temperature

information, exemplary power supply (e.g., DC power supply current share tracking). The information may be provided in line charts, such as with various data points charted in different colors, with different patterns, with different symbols, combination thereof, or the like. Information may be provided for a single unit, or multiple units, such as in individual charts or multiple charts. The type, kinds, and format of information illustrated, including what all is included in a given line chart, is merely exemplary and not intended to be limiting. The information may be plotted or otherwise provided over any length time period, such as but not limited to, 12 hours, 24 hours, 36 hours, etc. The information may be plotted or otherwise provided in substantially real time, or based on historical records. The information may be scaled, filtered, or the like.

[0064] The user display **90** of FIG. **20** includes exemplary proof-of-play information, such as in one or more reports, for an exemplary one of the display assemblies **30**, which may include, for example without limitation, display information, image information, plays delivered, campaign information, days played information, image thumbnails, combinations thereof, or the like.

[0065] The user display **90** of FIG. **21** includes exemplary lifetime and daily minimum and maximum operational parameter data displays for an exemplary one of the display assemblies **30**. The lifetime and/or daily minimum and maximum operational parameter data displays may be provided for any type and/or kind of data, including but not necessarily limited to, minimum lifetime and/or daily values for various data points, graphical displays of such information, combinations thereof, or the like.

[0066] The sensors may comprise, for example without limitation, one or more of temperature sensors, pressure sensors, ambient light sensors, airspeed sensors, power sensors (e.g., electric meters, voltage meters, current meters), combinations thereof, or the like. Any number, type, and/or kind of sensors may be provided for generating the various data points. The components **60** may include the sensors, though such is not required. The sensors may be in electronic communication with the controllers **50**. The sensors may be located at, or internal to, the display assemblies **30**. Alternatively, or additionally, the sensors may be external to and, remote from, the display assemblies **30**. For example, without limitation, some or all of the ambient conditions may be determined from one or more internet-based sources, which may include, for example without limitation, data obtained from one or more local weather sensors. Such data may be received at the controller **50**, such as by way of the network communication devices **40**.

[0067] Some or all of the user displays **90** may be accessible by way of one or more internet-based portals, webpages, applications, combinations thereof, or the like. The user displays **90** may be generated at access, upon navigation to particular ones of the user displays **90**, and/or various user selections.

[0068] The information displayed and the manner in which it is displayed is merely exemplary and is not intended to be limiting. Any type or kind of data may be displayed in any format.

[0069] Any embodiment of the present invention may include any of the optional or preferred features of the other embodiments of the present invention. The exemplary embodiments herein disclosed are not intended to be exhaustive or to unnecessarily limit the scope of the invention. The exemplary embodiments were chosen and described in order to explain the principles of the present invention so that others skilled in the art may practice the invention. Having shown and described exemplary embodiments of the present invention, those skilled in the art will realize that many variations and modifications may be made to the described invention. Many of those variations and modifications will provide the same result and fall within the spirit of the claimed invention. It is the intention, therefore, to limit the invention only as indicated by the scope of the claims.

[0070] Certain operations described herein may be performed by one or more electronic devices. Each electronic device may comprise one or more processors, electronic storage devices, executable software instructions, and the like configured to perform the operations described herein. The electronic devices may be general purpose computers or specialized computing device. The electronic devices may be personal computers, smartphone, tablets, databases, servers, or the

like. The electronic connections described herein may be accomplished by wired or wireless means. The electronic devices, including but not necessarily limited to the electronic storage devices, databases, or the like, may comprise and/or be configured to hold, solely non-transitory signals.

## Claims

1. A system for remotely monitoring digital signage, said system comprising: at least one display assembly, having components including at least one cooling fan, at least one electronic display, and at least one sensor adapted to monitor, in substantially real time, at least one operating condition at said at least one display assembly; a programmable controller adapted for communication over the internet, said controller in electronic communication with said at least one sensor and at least one of said components; at least one database having stored therein one or more preferred performance data settings for said at least one operating condition; at least one remote computer in electronic communication with said controller, said computer comprising software instructions, which when executed, configure the at least one remote computer to: receive and store, in substantially real time, data representative of actual performance of said at least one operating condition, from said at least one sensor; automatically analyze said actual performance data in comparison with said preferred performance data settings to determine if they differ by at least a preset threshold difference; and if said preset threshold difference is determined to exist, automatically send an electronic signal to said controller to adjust operations of at least one of said components to more closely align said actual performance data with said preferred performance data settings.
2. The system of claim 1, wherein: said at least one sensor comprises a temperature sensor; and said at least one operating condition is an internal operating temperature of said at least one display assembly.
3. The system of claim 1, wherein: said electronic signal is an electronic instruction to said at least one cooling fan, via said controller, to increase an operational speed level of said at least one cooling fan.
4. The system of claim 1, wherein: said electronic signal is an electronic instruction to said at least one electronic display, via said controller, to decrease an operational illumination level of said at least one electronic display.
5. The system of claim 1, wherein: said preset threshold difference is at least 5% different from said preferred performance data settings.
6. The system of claim 1, wherein: said at least one operating condition is ambient temperature in a same immediate vicinity of said at least one display assembly.
7. The system of claim 1, wherein: the electronic display includes a backlight and liquid crystals.
8. The system of claim 1, wherein: the programmable controller and the at least one database are provided at said at least one display assembly.
9. The system of claim 1, wherein: at least one of said one or more preferred performance data settings are time dependent on time of day at the location of said at least one display assembly.
10. The system of claim 1, wherein: said at least one sensor comprises a light sensor; and said data representative of actual performance is a luminance level of said electronic display.
11. The system of claim 1, wherein: said data representative of actual performance is an image of what is showing on said at least one electronic display in substantially real time.
12. The system of claim 11, wherein: said preset threshold difference is met if the electronic display is off.
13. The system of claim 12, wherein: said electronic signal is a restart signal sent to said electronic display.
14. The system of claim 1, wherein: said at least one sensor is adapted to sense whether a door has been left open on said at least one display assembly.

**15.** The system of claim 1, wherein: said at least one sensor is an accelerometer adapted to sense a physical impact incurred by said at least one display assembly, and wherein said preset threshold difference is a force at least 10% greater than all normal and customary forces experienced by said at least one display assembly.

**16.** The system of claim 1, wherein: said at least one sensor is a power sensor adapted to monitor electrical power used by said at least one display assembly.

**17.** The system of claim 1, wherein: said at least one remote computer is further configured to, where said preset threshold difference is determined to exist, generate and transmit an alert to at least one remote electronic device.

**18.** The system of claim 1, wherein: said at least one remote computer is further configured to cause generation of a visualization of said actual performance data at one or more remote electronic devices.

**19.** A system for remotely monitoring digital signage, said system comprising: at least one display assembly, having components including at least one cooling fan, at least one electronic display comprising a backlight and a liquid crystal layer, at least one sensor adapted to monitor, in substantially real time, at least one operating condition at said at least one display assembly including at least one temperature level; a programmable controller adapted for communication over the internet, said controller in electronic communication with said at least one sensor and at least some of said components, including the at least one cooling fan and the at least one electronic display ("monitored components"); at least one database having stored therein one or more preferred performance data settings for said at least one operating condition and said electronic display; at least one remote computer in electronic communication with said controller, said computer comprising software instructions, which when executed, configure the at least one remote computer to: receive and store, in substantially real time, data representative of actual performance of said at least one operating condition, from said at least one sensor, and current operating conditions of said monitored components; automatically analyze said actual performance data in comparison with said preferred performance data settings to determine if they differ by at least a preset threshold difference; and if said preset threshold difference is determined to exist, automatically send an electronic signal to said controller to adjust said actual performance of at least one of said monitored components based on said actual performance data to more closely align said at least one operating condition to said preferred performance data settings.

**20.** A system for remotely monitoring digital signage, said system comprising: at least one display assembly, each having components including cooling fans, an electronic display comprising a backlight and a liquid crystal layer, sensors adapted to monitor, in substantially real time, operating conditions including at least one ambient temperature level, at least one internal temperature level for the display assembly, and at least one ambient light level at the display assembly, a programmable controller adapted for communication over the internet, said controller in electronic communication with said sensors and operative electronic communication with said cooling fans, and the electronic display, and at least one database having stored therein predetermined preferred performance parameters for said operating conditions, said cooling fans, and said electronic display; at least one remote computer in electronic communication with said controller, said computer comprising software instructions, which when executed, configure the at least one remote computer to: receive and store, in substantially real time, data representative of actual performance of said operating conditions from said sensors, and current operating conditions of said electronic display and said cooling fans; automatically analyze said actual performance data in comparison with said predetermined preferred performance parameters to determine if data from any one or more of the sensors, the electronic display, and said cooling fans differ from a respective one of the predetermined preferred performance parameters by at least a preset threshold difference; and if said preset threshold difference is determined to exist with regard to any one or more of the sensors, the electronic display, and said cooling fans, automatically send an electronic signal to said

controller to adjust said actual performance of one or more of said components, including at least one of the cooling fans and the electronic display, thereby adjusting at least one of said operating conditions to more closely align with at least the respective one of said predetermined preferred performance parameters.

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